

## BATTERY MANAGEMENT SYSTEM FOR ELECTRIC VEHICLE

Sagar Sutar<sup>\*1</sup>, Ramkrushna Shinde<sup>\*2</sup>, Dhanashree Patil<sup>\*3</sup>, Sayali Jamdade<sup>\*4</sup>,  
Rutuja Chougule<sup>\*5</sup>, Prof. Sanjay Dhaygude<sup>\*6</sup>, Rohan Doshi<sup>\*7</sup>

<sup>\*1,2,3,4,5</sup>Student, Department Of Electronics Engineering, Walchand Collage Of Engineering,  
Sangli, Maharashtra, India.

<sup>\*6</sup>Training & Placement Officer (TPO), Project Guide, Department Of Electronics Engineering,  
Walchand Collage Of Engineering, Sangli, Maharashtra, India.

<sup>\*7</sup>Senior Technical Lead, Mentor, FORVIA HELLA India Automotive, Pune, India.

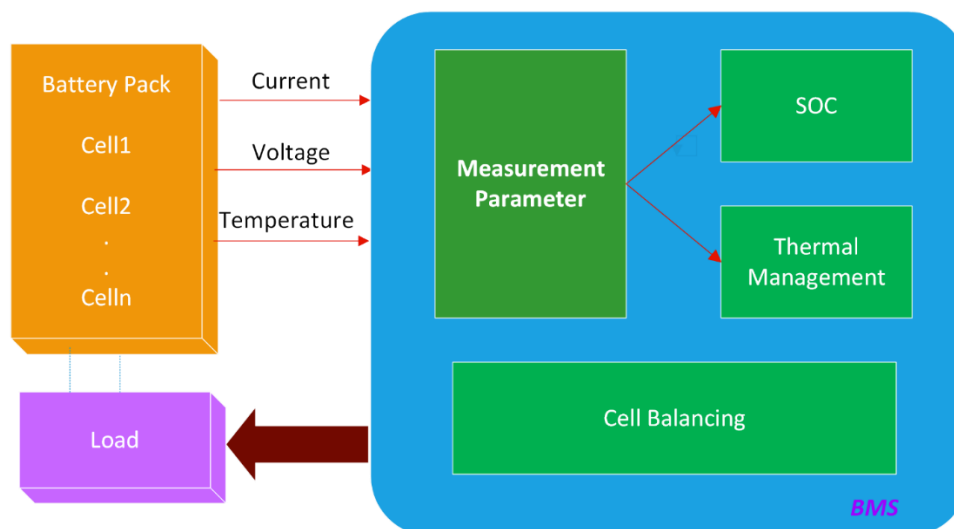
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### ABSTRACT

Battery Operation system / Battery management system (BMS) is one of the most important parts of any electric vehicle which is used to cover and control the charging and discharging of rechargeable batteries. Which ensures the safe operation of battery. It monitors the parameter determine soc & it provides necessary service to ensure safe operation of battery. In order to maintain the state of the battery current, voltage and ambient temperature, different monitoring techniques are used. For monitoring purpose various sensors are used with microcontrollers this paper addresses state of charge, temperature monitoring & control, cell balancing. By considering all these concepts future challenges and possible solutions can be obtained.

**Keywords:** SOC (State Of Charge), Temperature Control, BMS (Battery Management System) Etc.

### I. INTRODUCTION



**Fig 1.** Block Diagram of BMS

Electric vehicles are the future of transportation. EVs are playing a crucial part because of its zero emigration of dangerous feasts and use of effective energy. Electric vehicles are equipped by a large number of battery cells which requires an effective BMS while they're furnishing necessary power.

Battery Management system (BMS) is the pivotal system in electric vehicle because batteries used in electric vehicle should not be get overcharged or over discharged. If that happens, it leads to the damage of the battery, rise in temperature, reducing the life span of the battery, and occasionally also to the persons using it. It's also used to maximize the range of vehicle by duly using the quantum of energy stored in it.

Battery Management system is essential for following reasons:

1. Maintain the safety and the trustability of the battery.
2. Battery sate monitoring and evaluation.
3. To control the state of charge.

4. For cell balancing and temperature controlling.

5. operation of regenerative energy.

## II. METHODOLOGY

### SOC (State of Charge):

The soc measurement of battery is an important parameter or feature of the battery capacity. it can't be measure directly. it can be only estimated through measurable variables such as current, voltage and temperature. Various SOC estimation methods are there but Accurate soc (state of charge) estimation is one of the main issues in BMS. an equivalent electric circuit has been used to describe the battery model.

State of charge is defined as the available quantum of battery as the chance of rated capacity of the battery. State of charge gives a vital support to battery operation system to assess the state of the battery which helps the battery to operate within the safe operating range by controlling charging and discharging. The battery lifetime is also increased by this. State of charge can't be estimated directly. It's calculated by using the equation

$$SOC = 1 - \frac{\int idt}{C_n} \dots\dots\dots(1)$$

Where I =current and

Cn= maximum capacity that the battery can hold

There are multiple methods to estimate the state of charge. Following are the list of state of charge estimation system:

1. OCV SOC estimation system (Open Circuit Voltage)
2. Coulomb counting SOC estimation system
3. Kalman filtering SOC estimation system

Among all these different methods Kalman filtering system has been successful for the estimation of SOC for EV'S.

### Temperature Monitoring & Control:

During operations of battery because of chemical reactions it produces heat in batteries. as load varies there is chances of temperature rise or drop, damage the chemical property of cells and it can lead to explosion in worst cases.

The temperature monitoring and control is essential for safer and smooth operation of batteries. The higher temperature of batteries can lead abnormal behavior like fire catch up and lower temperature can affect the charging and discharging parameter (current or voltage) of battery, may it can also reduce the power handling capability of batteries.

### Cell Balancing:

In order to meet the requirements of energy storage rechargeable batteries are widely used in electric vehicle and other areas also. in various applications because of low terminal voltage of individual battery cells, they all are normally connected in series to create battery pack to reach required voltage level. however, the imbalances between individual cells of a battery pack are a common phenomenon. a well- known imbalances is SOC (State of charge) differences between all cells [4] This imbalance is caused by both intrinsic and extrinsic difference among battery cell [5]

The soc differences between cells are because of energy consumption of each cell is different. A li-ion battery pack is having safety issue if overcharging and undercharging occurs, at the time of discharging the cell with lowest soc will reach the lower limit of safe operation voltage first thus the BMS has to turn of the operation. Same things (similar situation) happen during charging of battery. the cell with highest soc will reach upper limit of safe operation voltage first.

This overcharge and undercharge cause on SOH (state of health) of battery. In order to maintain a balanced battery bank two methods are there: 1. Passive cell balancing 2. Active cell balancing

### III. MODELING AND ANALYSIS

State of Charge & Thermal management:

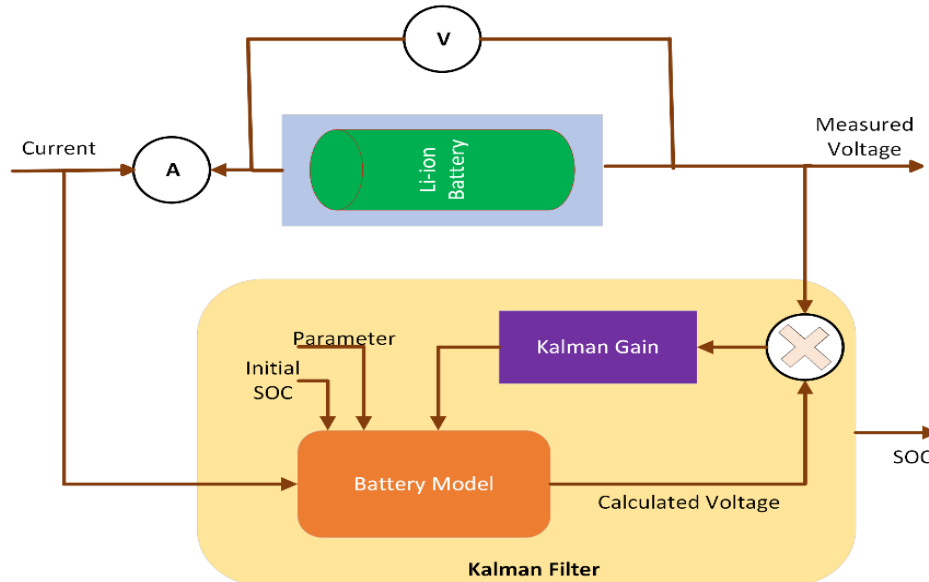


Fig 2. Block Diagram of Kalman Filter

The SOC is one of the most important information in a BMS; it is an inner state of each cell [6], which cannot be measured directly during operation of the battery. Accordingly, estimation is the only way to determine a value of SOC. SOC is mathematically defined as follows [7]:

$$SOC_{batt} = SOC_{initial} - \frac{1}{3600 \times Ah} \int I_{batt} dt \quad \dots\dots\dots(2)$$

$SOC_{batt}$ : The state of charge of the battery.

$SOC_{initial}$ : The initial state of charge.

Ah: Battery capacity in A / H.

$I_{batt}$ : Battery current.

Under the assumption that all the parameters of the model are constant, we have linear equations structuring our model; therefore, we can use a linear state observer such as Kalman filter algorithm. This observer equations are given below and will be used for our next presented results. (Fig 4,5&6).

#### A. SOC estimation using Kalman filter:

The Kalman filter algorithm can be represented in four equations as follows:

Prediction of state space variables

$$x_k = A\hat{x}_{k-1} + Bu_{k-1} \quad \dots\dots\dots(3)$$

Prediction of the covariance matrix

$$P_k = A\hat{P}_{k-1}A^T + Q \quad \dots\dots\dots(4)$$

Update of the Kalman gain

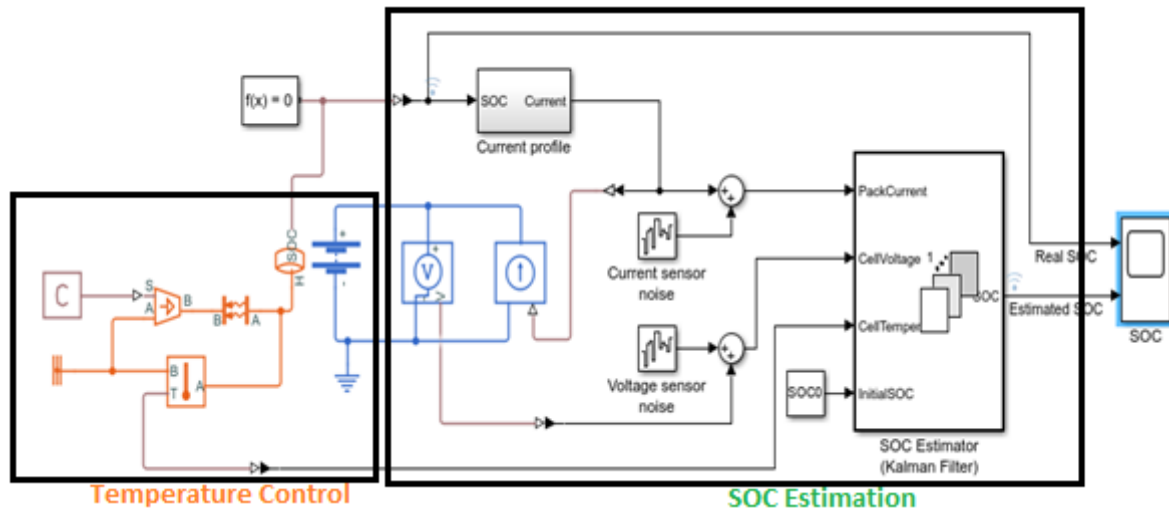
$$K = P_k C_k^T [C_k P_k C_k^T + R]^{-1} \quad \dots\dots\dots(5)$$

Estimation of the state variable and correction of the prediction

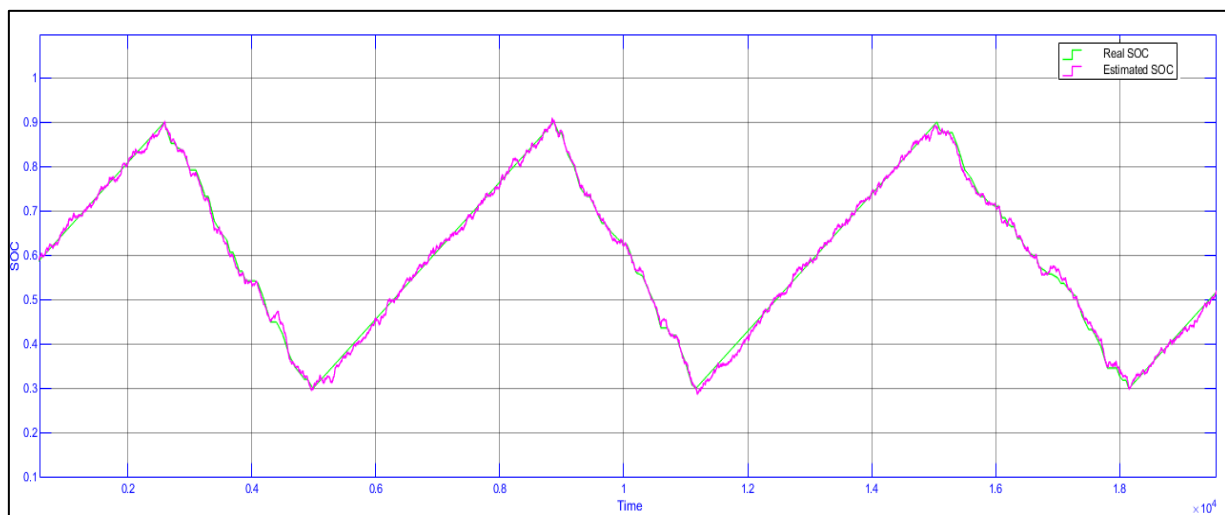
$$\hat{x}_k = x_k + K[y_k - C_k x_k - D_k u_k] \quad \dots\dots\dots(6)$$

Estimation of the estimation error

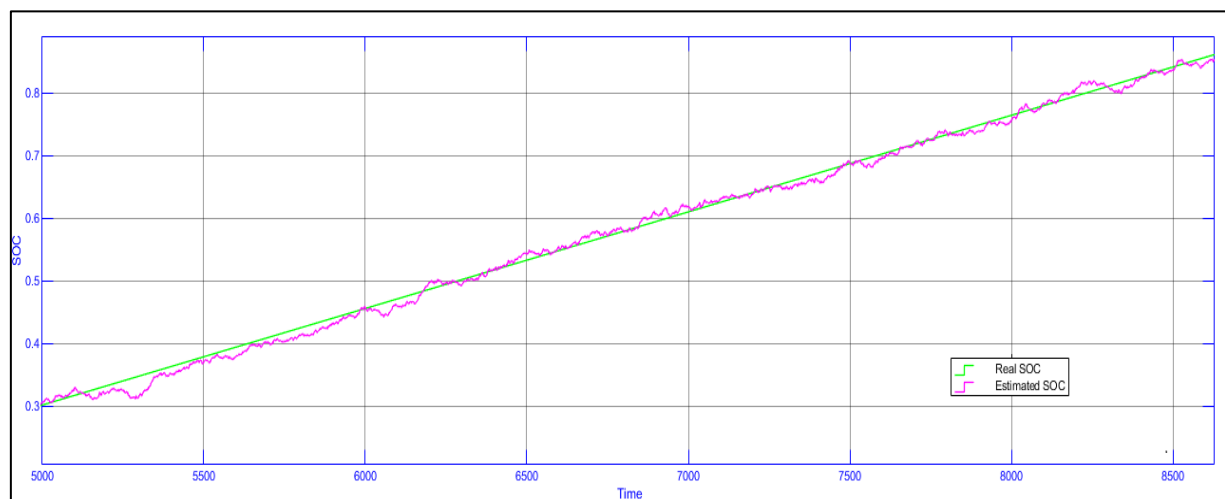
$$\hat{P}_k = (I - KC_k)P_k \quad \dots\dots\dots(7)$$



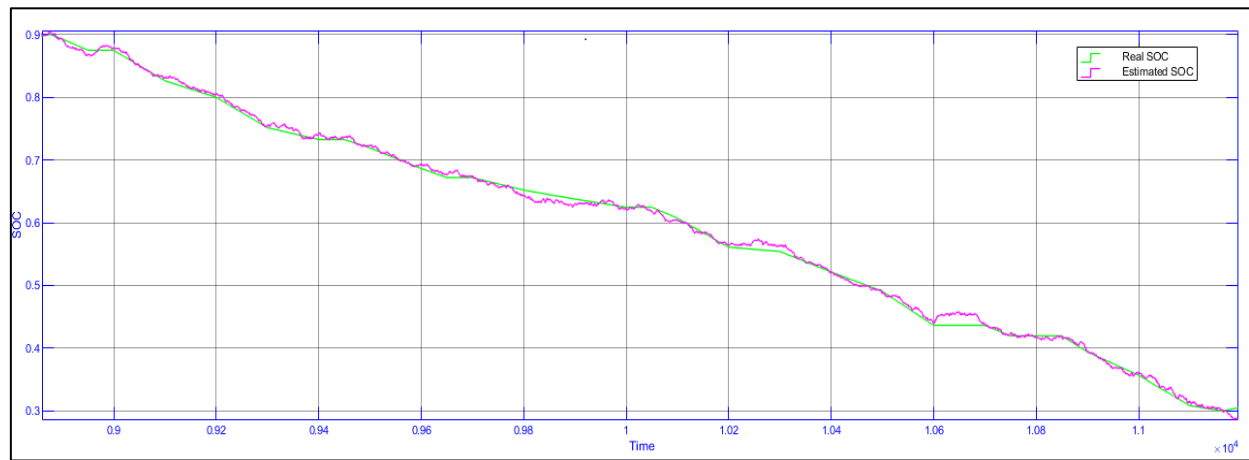
**Fig 3.** Circuit Diagram of SOC Estimation and Thermal management.



**Fig 4.** Difference between Real & Estimated SOC.

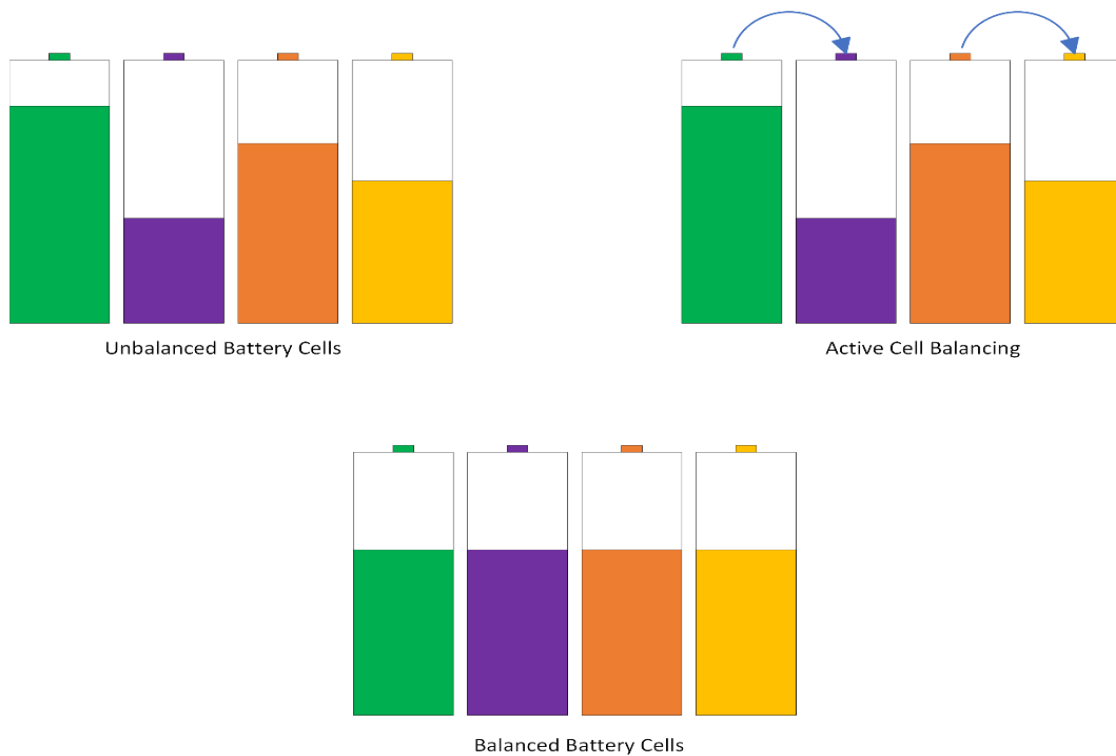


**Fig 5.** Charging



**Fig 6. Discharging**

### Cell Balancing:

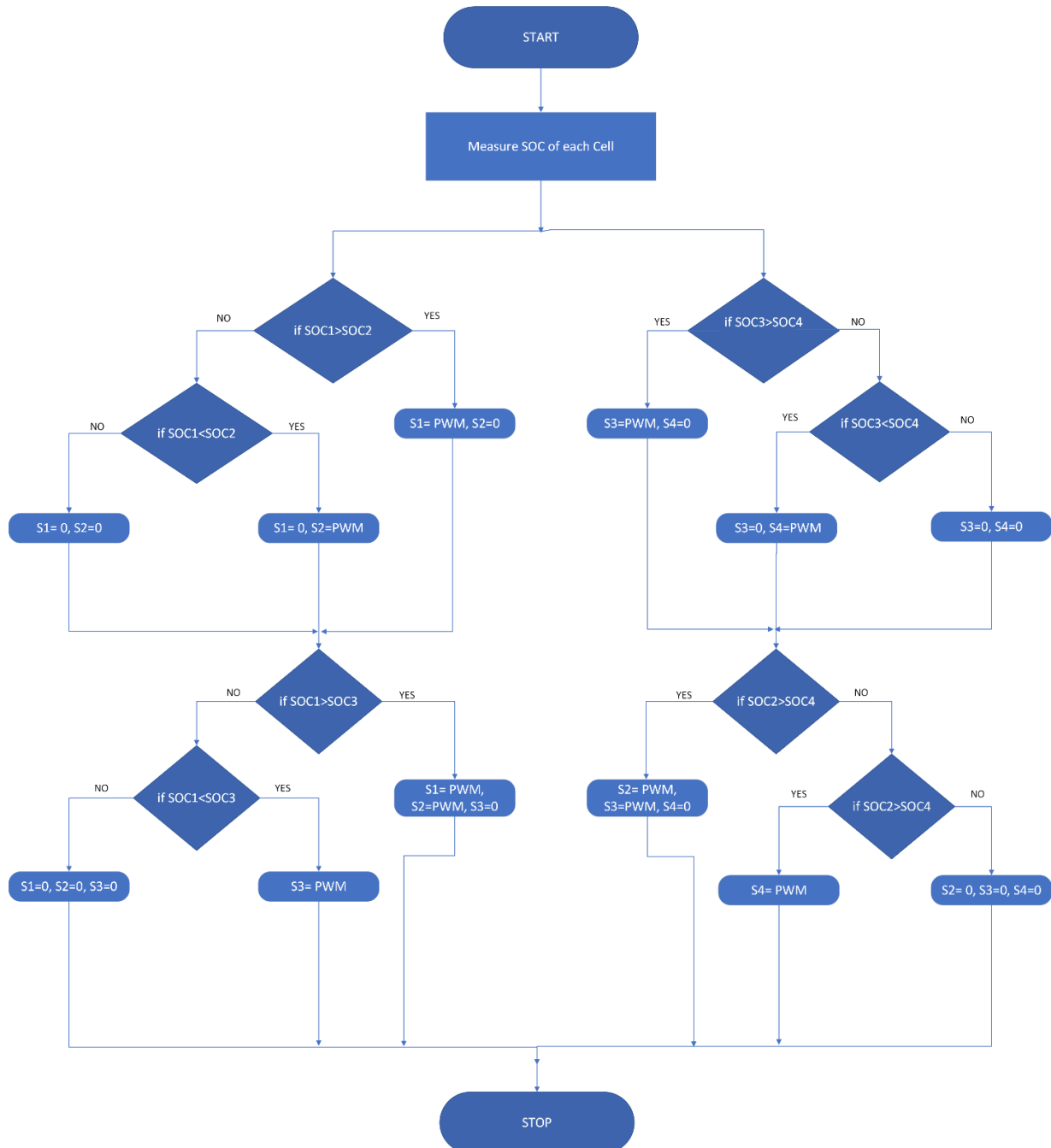


**Fig 7. Cell Balancing**

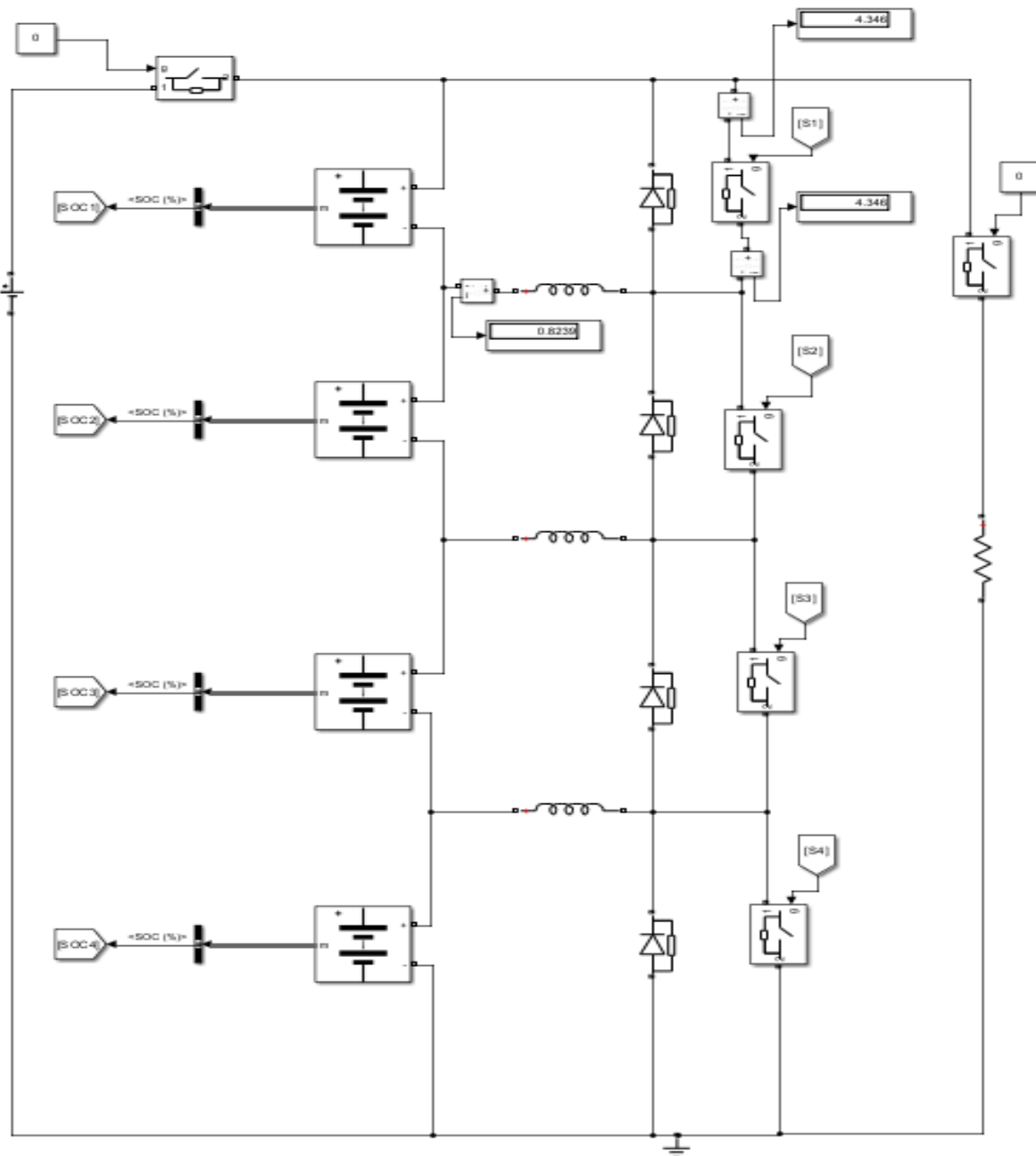
Active cell balancing is a point set up in some battery Management systems (BMS) that helps to keep the voltage of all the cells in a lithium- ion battery pack within safe situations. This is done by covering the voltage of each cell and, when necessary, transferring energy from the advanced- voltage cells to the lower- voltage cells. By keeping all the cells within a certain voltage range, it helps to help anyone cell from getting overcharged or discharged, which can dock the overall lifetime of the battery pack.

There are numerous different ways that active cell balancing can be enforced, but utmost systems use some form of MOSFETs (essence- oxide- semiconductor field- effect transistors) to control the inflow of current between cells. When a cell's voltage gets too high, the current is diverted down from that cell and into another cell with a lower voltage. This process continues until all the cells in the battery pack are within the safe voltage range. While active cell balancing can help to protract the life of a lithium- ion battery pack, it's not reliable and there are still some pitfalls associated with using this type of system. One implicit issue is that if one cell becomes damaged or demoralized, it can beget problems for the rest of the cells in the pack. also, if the BMS

itself fails, it could lead to disastrous failure of the entire battery pack. Active cell balancing is a point in battery operation systems that helps to keep the cells in a lithium- ion battery pack within a safe operating range. This is done by periodically covering the cell voltages and currents and conforming the charge and discharge rates consequently. By doing this, active cell balancing can help to help unseasonable aging of the cells and extend the overall life of the battery pack.

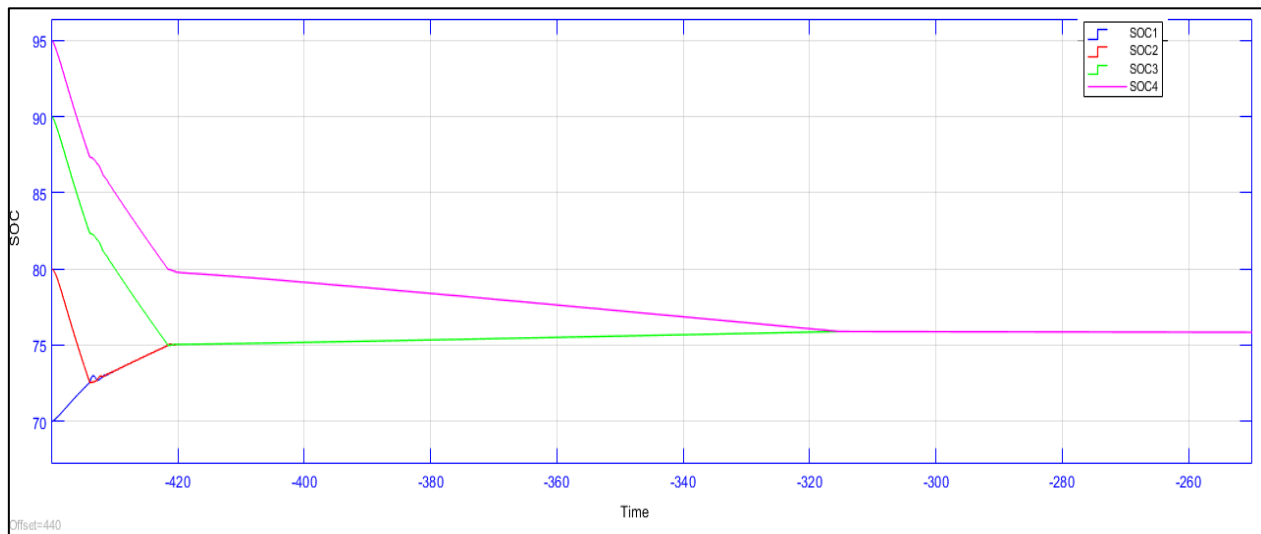


**Fig 8.** CB Flowchart



**Fig 9. Cell Balancing Circuit Diagram.**

In This first SOC of cell1 and cell2 will be compared if SOC1 is greater than SOC2 then switch (s1) will turn ON and energy will transfer from cell1 to cell2 through switch and inductor. If SOC2 >SOC1 then switch (s2) will turn ON and energy will transfer from cell2 to cell1 through diode1. after that SOC of cell3 and cell4 will be compared if SOC3 is greater than SOC4 then switch (s3) will turn ON and energy will transfer from cell3 to cell4 through switch & inductor. If SOC4 >SOC3 then switch (s4) will turn ON and energy will transfer from cell4 to cell3 through diode3. Next SOC of cell1 and cell3 will be compared if SOC1 is greater than SOC3 then switch (s1) & switch (s2) will turn ON and energy will transfer from cell1 to cell3 through switch and inductor. If SOC3 >SOC1 then switch (s3) will turn ON and energy will transfer from cell3 to cell1 through diode2 & diode1. Next SOC of cell2 and cell4 will be compared if SOC2 is greater than SOC4 then switch (s2) & switch (s3) will turn ON and energy will transfer from cell2 to cell4 through switch and inductor. If SOC4 >SOC2 then switch (s4) will turn ON and energy will transfer from cell4 to cell2 through diode3 & diode2. Same process will follow for 'n' number of cells.



**Fig 10. Cell Balancing Output**

#### IV. RESULTS AND DISCUSSION

The model of the complete soc estimation and temperature monitoring and control in fig. 3 has been developed and tested with noisy signal. the fig.3 consists of two parts 1. Temperature control 2. SOC estimation using Kalman filter. 1). When battery is in operation, the electrochemical processes which produces heat which increase the battery temperature for that convective heat transfer is used to maintain temperature of battery. 2. For soc estimation Kalman filter method is used this method is one of most accurate soc estimate method.

In fig4 the waveform indicates the difference between real soc and estimated soc which are approximately equal (Green color signal indicates real soc & pink color signal indicate estimated soc). Fig10 shows the output of cell balancing circuit the technique of cell balancing used in this paper is active cell Balancing. initial SOC of each cell is different in fig10 and after some time the SOC of all cells are maintained at constant level.

#### V. CONCLUSION

In this way we are developing the model for battery management system in electric vehicle by controlling key parameters like current, voltage, temperature, and SOC (State of Charge).

It is an important that the BMS should be well maintained with battery credibility & security. this paper mainly focusses on the study of battery management system and enhance the power performances of electric vehicles. Besides, the goal of reducing the greenhouse gas can greatly be achieved by using battery management system.

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